KHALID LAMRINI

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EDUCATION

ENS Paris-Saclay || Paris, France *September 2026 – April 2027*
M2 MVA (Mathematics, Vision and Learning)

- Research master's in mathematics for machine learning and computer vision.

Ecole Polytechnique || Paris, France September 2023 – April 2026
Master's Degree in Engineering

- France 's leading engineering school (awarded a merit scholarship by the Moroccan government)
- Coursework focused on applied mathematics, machine learning, deep learning, stochastic modeling, optimization, signal processing, and algorithms.
- 6-month research project under the supervision of Ziad Kobeissi: "Resolution of Fisher-KPP type reaction–diffusion equations through learning-based methods (PINNs), with applications to tumor proliferation".
- Multimodal generative AI project on food nutrition estimation from images and textual information.
- Developed a web application to calculate Elo scores for competitive student games, including account management and server-side logic.

Collège Stanislas || Paris, France September 2021 – June 2023
Exam Preparation Classes

- Intensive coursework in Mathematics, Physics and Chemistry, preparing for competitive entrance exams for top French engineering schools and universities.

EXPERIENCE

Columbia University: *Visiting Researcher* April 2026 – August 2026

- Conducted research at the Biomedical Informatics Lab, Columbia University, supervised by Prof. Shalmali Joshi.
- Psychiatric Hospitalization Risk Prediction: Developed linear and deep learning methods to predict risk using All of Us (AoU) genomic data. Configured computing clusters and built data processing pipelines using Hail and PLINK to handle large-scale datasets. Analyzed model sensitivity to SNP counts across diverse populations.
- Neuron Conductance Prediction: Implemented a reduced Hodgkin-Huxley (HH) model to predict electrophysiology for within-type conductances over mouse neurons. Trained a transcriptomics-based autoencoder to evaluate predictive performance improvements compared to type-only baseline models.

Saint-Gobain Glass France: *Data Scientist Intern* June 2025 – August 2025

- Built a pilot outlier detection tool for the coating line process, covering the full pipeline from data cleaning and preprocessing to correlation analysis, statistical modeling, and validation.
- Configured and deployed Databricks Genie spaces, providing Saint-Gobain teams with natural language access to databases and creating best practices to facilitate adoption.

COMPUTER SKILLS

- **Programming Languages:** Python (advanced), Java, HTML, SQL, PHP
- **Data Science**: Scikit-learn, PyTorch, TensorFlow, Pandas, Polars

EXTRACURRICULAR

- Active member of the Gastronomy Association and the International Students' Association
- Sports: Tennis, Running, Football
- Native in French and Arabic and proficient in English (C1-C2 level in TOEFL iBT)